The lecture moves on.
Your understanding should too.

An EEG-assisted learning companion that helps students revisit and understand the moments they may have missed.

Catch the gap. Get back to learning.

YouTube lectures
+ local AI recovery

A small lapse can create a bigger learning gap.

01

Lose the thread

One missed step makes the next idea harder to follow.

02

Hunt for the moment

Scrubbing backward means guessing what was missed.

03

Replay without clarity

Hearing it again may still leave the question unanswered.

The opportunity: turn a moment of confusion into a moment of recovery.

From possible distraction to a useful next step.

SENSE

Notice the moment

EEG signals indicate a possible attention lapse.

CAPTURE

Save the context

The extension records the lecture timestamps.

RECOVER

Choose your help

Replay the segment or ask AI to explain it.

Student stays in control: replay, explain, or continue.

Turn a brain signal into a useful learning action.

01 / ACQUIRE

NeuroPawn board
EEG via LSL

02 / FILTER

Alpha: 8–13 Hz
Beta: 13–30 Hz

03 / COMPARE

Band-power ratio
vs personal baseline

04 / RESPOND

Save video interval
Replay or explain

RECOVERY: video timestamps → overlapping captions → local Ollama

The language model explains the lecture; it does not classify the EEG.

Implemented pipeline. Focus classification is a rule-based prototype, not a trained neural network.

A simple signal we can explain and calibrate for each student.

OUR FEATURE

R = alpha power / beta power

Higher relative alpha is our candidate signal of reduced engagement.

Research-informed starting point

Related studies evaluate beta/alpha.
We use its inverse as an alert feature.

Practical for this prototype

Low computation. No labeled training set.
Compared with a personal baseline.

A testable heuristic—not proof that a student stopped paying attention.

Research: Coelli et al., 2015, DOI 10.1109/EMBC.2015.7318658. Sources and limitations in backup slides.

Less searching. More purposeful review.

ILLUSTRATIVE RECOVERY CARD

Neurons · 02:10–02:35

Dendrites receive input.
The axon carries signals to other cells.

Quick check: What does an axon do?

TARGETED

Review the relevant moment, not a whole lecture.

ON THE STUDENT'S TERMS

Student-controlled recovery.
Local AI by default.
History they can clear.

Working prototype: interval capture, replay, AI explanations, and saved history. Learning impact remains to be tested.

Start with one course. Grow through campuses.

FIRST USERS

College students revising recorded STEM lectures

FIRST BUYERS

Learning-support teams and teaching departments

PROPOSED REVENUE

Campus software licenses, after pilot validation

Lower the adoption barrier: start with a small pool of loaner headsets.

Help us turn a working prototype into a better study experience.

PROPOSED PILOT

1
course partner

20
student volunteers

2
weeks of use

Measure comprehension, recovery time, and willingness to keep using it.

NeuroLearn. Recover the moment. Keep the momentum.

The rule behind the attention signal.

EEG WINDOW

5 seconds

First incoming channel; demeaned. Butterworth filters + filtfilt.

CALIBRATION

10 seconds

Mean alpha/beta ratio during focused calibration.

ALERT RULE

1.8× baseline

Above threshold continuously for 5 seconds → alert.

$$P = \text{mean}(\text{filtered signal}^2) \qquad R = P_{\text{alpha}} / (P_{\text{beta}} + 10^{-9})$$

Pending starts interval tracking immediately; the alert waits 5 s. Both end when focus returns.

Source: current test.py. The multiplier and durations are prototype settings, not clinically validated thresholds.

Why this feature—and what comes next?

WHY WE STARTED HERE

Interpretable and cheap to compute.
Beta/alpha has been studied in attention.
Its inverse gives a convenient alert direction.
A personal baseline gives a local reference.

WHAT WE NEED TO VALIDATE

Sensitivity to task, electrode site, and artifacts.
False alerts in real lecture sessions.
Compare beta/(alpha + theta) and classifiers.
Measure against labeled attention checks.

Research supports investigating EEG ratios—not our exact threshold or learning outcomes.

Coelli et al. (2015): beta/alpha during sustained attention; recording sites mattered.
Ahmed et al. (2025): tested several indices; combining features outperformed single indices in their task.

References: doi.org/10.1109/EMBC.2015.7318658 | doi.org/10.3390/s25072072
No detection-accuracy claim is made for NeuroLearn.